

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
ELECTRICAL ENGINEERING DEPARTMENT

Class Test-1 (Session 2025-26) (SET-1)

Course: B. Tech

Subject: Fundamentals of Electrical Engineering

Subject Code: BEE-101

Semester: Ist

Section: E, F, G, H

Time: 60 Minutes

Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO2	CO2	CO2	CO1	CO1	CO3
Bloom's Level	K1	K3	K2	K3	K3	K2

Note: 1) This paper contains three sections, Section (A) (B) & (C)

2) All sections are compulsory

Section (A)

Attempt this question. This question carries 2 marks.

(1*2 = 2)

- 1) Derive expression for resonant frequency in series R-L-C circuit.

Section (B)

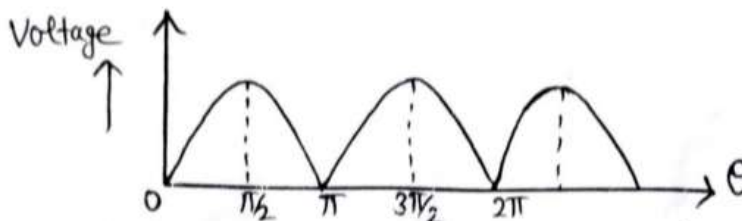
Attempt all questions. Each question carries 3 marks.

(2*3 = 6)

- 2) An alternating voltage is given by $v = 141.4 \sin 314 t$. Find (i) the instantaneous value of voltage when 't' is 3 ms (ii) the time taken for the voltage to reach 100 V for the first time after passing through zero value.

OR

Find rms average value and peak factor of the voltage waveform shown in figure.



- 3) For an ac circuit $v = 200 \sin 377 t$ volts, and $i = 9 \sin (377 t - 30^\circ)$ amperes. Determine true power, apparent power and reactive power of circuit.

OR

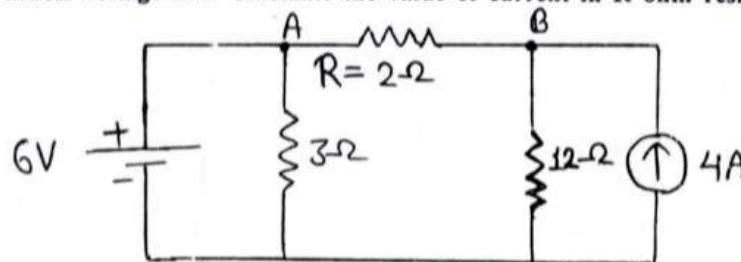
Show that the average power for purely inductive circuit is zero.

Section (C)

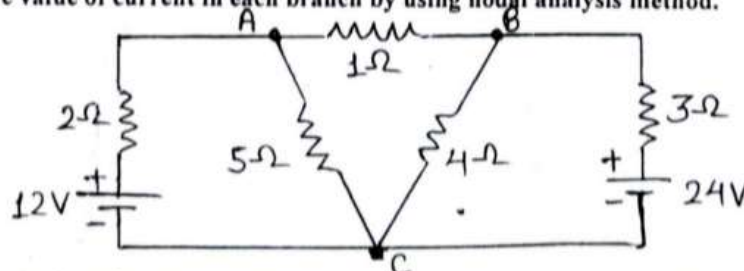
Attempt all questions. Each question carries 4 marks.

(3*4 = 12)

- 4) State Kirchhoff voltage law. Calculate the value of current in R ohm resistance by using mesh analysis method.



- 5) Define with examples (i) Unilateral and Bilateral Elements.
 (ii) Determine the value of current in each branch by using nodal analysis method.



- 6) (i) Give the names (write only names) of different types of magnetic materials with examples.
 (ii) Derive emf equation of normal two winding transformer.

approved

Saurabh
08/10/25

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
ELECTRICAL ENGINEERING DEPARTMENT
Class Test-1 (Session 2024-25) (SET-1)

Course: B. Tech

Subject: Fundamentals of Electrical Engineering

Subject Code: BEE-201

Semester: 1st

Section: A, B, C, D

Time: 60 Minutes

Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO3	CO2	CO2	CO1	CO1	CO2
Bloom's Level	K1	K2	K3	K3	K3	K4

Note: 1) This paper contains three sections, Section (A) (B) & (C)

2) All sections are compulsory

Section (A)

(1*2 = 2)

Attempt this question. This question carries 2 marks.

- 1) With the help of examples classified different types of magnetic materials.

Section (B)

(2*3 = 6)

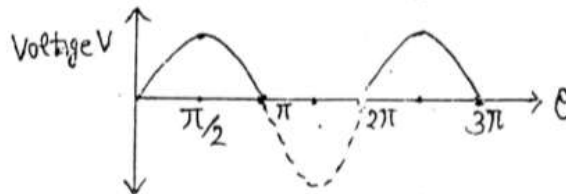
Attempt all questions. Each question carries 3 marks.

- 2) Derive an expression for form factor and peak factor for sinusoidal voltage.

OR

Find rms, average value and peak factor of the voltage waveform shown in figure.

Peak factor = $\frac{\text{max}}{\text{rms}}$



- 3) The voltage applied to a circuit is $v = 100 \sin(\omega t + 30^\circ)$ & current flowing in the circuit is $i = 20 \sin(\omega t + 60^\circ)$. Determine the impedance, resistance, reactance, power and power factor of the circuit.

OR

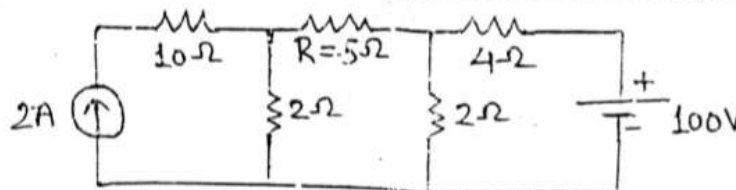
An alternating voltage is given by $v = 141.4 \sin 314 t$. Find (i) frequency (ii) rms value (iii) average value (iv) the instantaneous value of voltage when 't' is 2 ms (v) the time taken for the voltage to reach 100 V for the first time after passing through zero value.

Section (C)

(3*4 = 12)

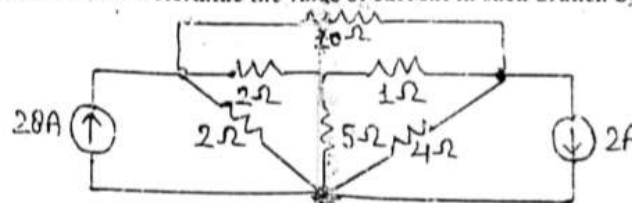
Attempt all questions. Each question carries 4 marks.

- 4) (a) Define with examples (i) Unilateral and Bilateral Elements (ii) Active and Passive Elements
 (b) State Kirchhoff voltage law. Calculate the value of current in R ohm resistance by using mesh analysis method.



Resistor network circuit

- 5) State Kirchhoff current law. Determine the value of current in each branch by using nodal analysis method.



- 6) Define the term Resonance. Derive expression for resonant frequency in series R-L-C circuit. Also draw resonance curve.

For any query contact to: Mr. Saurabh Saxena 7895043193, Mr. Akash Kr. Gautam 8954751891, Mr. Lalit Pareekha 7060460472

Approved
 Sachin
 17-05-25

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
ELECTRICAL ENGINEERING DEPARTMENT

Course: B. Tech

Subject: Fundamentals of Electrical Engineering

Subject Code: BEE-101

Class Test-2 (Session 2025-26) (SET-1)

Semester: 1st

Section: E,F,G,H

Time: 60 Minutes

Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO4	CO4	CO4	CO3	CO5	CO5
Bloom's Level	K1	K3	K3	K3	K2	K3

Note:

1) This paper contains three sections, Section (A) (B) & (C)

2) All sections are compulsory

Section (A)

Attempt this question. This question carries 2 marks.

(1*2 = 2)

Q1) What is the role of commutator and brushes in DC machine.

OR

Derive the formula for EMF generated in DC machine.

Section (B)

Attempt all questions. Each question carries 3 marks.

(2*3 = 6)

Q2) A 4-pole, long shunt, lap wound generator supplies 25 kW at a terminal voltage of 500 V. The armature resistance is 0.03 ohm, series field resistance is 0.04 ohm and shunt field resistance is 200 ohm. The brush drop may be taken as 1 V. Determine the emf generated.

Q3) Why single phase induction motor is not self starting? Also explain working of shaded pole motor.

OR

Explain the working principle of 3-phase induction motor.

A 4-pole, 3-phase, 50 Hz induction motor has a full-load slip of 4%. Calculate full load speed of motor.

OR

Explain the working principle of synchronous motor

Section (C)

Attempt all questions. Each question carries 4 marks.

(3*4 = 12)

Q4) In a 25 kVA, 2000 V/200 V transformer the iron and copper losses are 200 W and 400 W respectively. Calculate the efficiency at half-full load and 0.8 power factor lagging. Determine also the maximum efficiency and the corresponding load.

OR

Define the term voltage regulation in transformer. Determine voltage regulation of transformer whose ohmic drop is 2.25 % and leakage reactance drop is 7.5 % at 0.8 of lagging.

Q5) (a) What is the need of earthing in electrical installation? Write the names of different methods of earthing.

(b) Write 2 advantages & 2 disadvantages of XLPE cable over PVC cables.

OR

Complete equations of Lead Acid & Nickel Iron batteries.

Q6) (a) A battery has taken a charging current of 5.2 A for 24 hours at a voltage of 1.8 V. while discharging it gave a current of 4.5 A for 24 hours at an average voltage of 1.48V. Calculate the ampere-hour efficiency of the battery.

(b) Write a short note on MCB.

For any query contact to: Mr Saurabh Saxena 9456031300/7895043195, Mr. Tanuj Kr. Bisht 9149273074

approved
Saurabh
 20.11.25

Rajin (7)

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD
ELECTRICAL ENGINEERING DEPARTMENT

Class Test-2 (Session 2024-25) (SET-1)

Course: B. Tech
Subject: Fundamentals of Electrical Engineering
Subject Code: BEE-201

Semester: 2nd

Section: A,B,C,D
Time: 60 Minutes
Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO3	CO5	CO4	CO3	CO4	CO5
Bloom's Level	K2	K2	K2	K3	K3	K3

Note:

1) This paper contains three sections, Section (A) (B) & (C)

2) All sections are compulsory

Section (A)

Attempt this question. This question carries 2 marks.

(1*2 = 2)

Q1) What are the different types of losses in transformer?

Section (B)

Attempt all questions. Each question carries 3 marks.

(2*3 = 6)

Q2) Write a short notes on ELCB & MCB.

OR

Write advantages of PVC cable and XLPE cable over other cables.

Q3) Draw torque slip characteristics of 3 phase induction motor. A 3 phase, 4 pole 50 Hz induction motor has a full-load slip of 4%. Calculate full-load speed of motor:

OR

Explain the working principle of Synchronous motor. Also write applications of synchronous motor.

Section (C)

Attempt all questions. Each question carries 4 marks.

(3*4 = 12)

Q4) (a) The maximum efficiency of a 100 kVA transformer is 98.40% and operates at 90% full load unity power factor. Calculate the efficiency of a transformer at unity power factor at full load.

OR

Derive the formula for emf equation of normal 2 winding Transformer. Also write working principle of transformer.

Q5) (a) A 20 kW, 200 V shunt generator has an armature resistance of 0.05 ohm and a shunt field resistance of 200 ohm. Calculate the power developed in the armature when it delivers rated output.

OR

Derive EMF equation of DC machine.

(b) Derive the formula for torque developed in DC machine.

OR

Explain double revolving field theory. Write the names of different starting methods of 1-phase I.M.

Q6) What is the need of earthing in electrical installation? Write short notes on pipe earthing.

OR

Explain working of Lead Acid & Nickel Iron batteries by writing Complete equations mentioning active materials.

For any query contact to: Mr Saurabh Saxena 9456031300/7895043195

Pipe Earthing
Rod Earthing

approved

(i) Saurabh
19.06.25

magnetic field Rotar
constant speed.
efficiency.